

Pondering the Public Internet

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Relating the insights of many collaborators...

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...and pre-meeting participants

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Note To Self.....

- Do not give talk following Dave Maltz...
- You'll look old, slow, and dumb

What This Talk Is Not

- Not about subtleties or technical details
 - Addressing a higher-level question
- Not describing a pre-meeting consensus
 - My personal synthesis, then context from group
- Not about new functionality
 - This is about ecosystem change
 - Which does lead to a technical research agenda
- This agenda not optimized for adoption
 - But for expected impact, which is:
prob. of adoption X impact of adoption

What This Talk Is

- Reflection on the fate of the public Internet
- Start by recalling the “classic Internet”...
- ...then compare it to the current Internet...
- ...then present a broad research agenda to reconcile the two, achieving the best of both

The “Classic” Internet

- End-to-end delivery via *interconnection*
 - **Recursive delivery of packets**
 - From source ISP to transit ISP(s) to dest ISP
- Interconnection relies on standards
 - IP and BGP
- Why does interconnection matter?
 - ***Global delivery without global infrastructures***
 - Can add coverage incrementally via new ISPs
 - Improves both *coverage* and *competition*

The “Classic” Economic Model

- Users “pay for access”
 - Access for both sending and receiving
- Money follows chain of connectivity
 - From both sender and receiver
- Why does this matter?
 - Easy to connect (pay a single provider)
 - Money can flow to the many other involved ISPs
- In short: ***pay locally, get service globally***

Call This “The Public Internet”

- Platform providing any-to-any connectivity
- Provides “level playing field” for applications
- Yes, the ISPs are commercial
- Public: because anyone can buy access
 - “a business open to the public”

What About The Current Internet?

- Two important (and related) trends
- “Edge-Enhanced” service
 - Content-delivery infrastructures (CDIs)
 - But want to think more generally: enhancements
- “The Death of Transit?” (Huston, 2016)
 - Growth of large private networks (many-to-one)
 - Most traffic goes directly between client ISP and content provider’s private network or CDI
- ***Better service, but different ecosystem***

Two Key Ecosystem Differences

- CDIs paid for by content-providers
 - Pay: deployment costs, or paying third-party
 - Enhancements deployed in many locations close to clients, far from the content provider
- CDIs not “interconnected”
 - Yes, they connect at IP level, but they don’t “peer” at enhancement level
 - Can’t incrementally add coverage via peering interconnections between CDIs

As A Result...

- ***Global service requires global deployments***
- CDIs are deployed in parallel at huge scale
 - Provides a single point of payment
 - High barrier to entry to build a CDI
- Lagging on two factors vs public Internet:
 - Coverage
 - Competition

Where's The Harm?

- Short term:
 - Market is extremely concentrated
 - Dominated by hypergiants
 - Increases the digital divide
- Long term:
 - Public Internet eventually plays minimal role
 - Just hypergiants and a bunch of last-mile networks
 - No longer a level playing field for innovation

We Face A Fundamental Challenge

- Can we interconnect edge enhancements?
 - Interconnection: better ecosystem
 - *Global service without global infrastructures*
 - Enhancements: better performance
 - *Necessary for today's workloads*
- Can we build an enhanced public Internet?
 - Provide better competition and coverage
 - Restore/retain “level playing field” for innovation
 - Retain better performance

What Is Needed To Meet Challenge?

- Standardization of enhancements
 - *The equivalent of IP*
- Method of interconnection (peering)
 - *The functional contract*
- Economic model (flow of money)
 - *The financial contract*
 - *Need simple payments yet achieve global scale*
- Extensibility yet uniformity of enhancements
 - *Perhaps the greatest challenge going forward*
 - *Something the public Internet utterly failed at....*

Sketch Of One Possible Approach

- Standardization of enhancements
 - **Open-source code**
- Method of interconnection
 - **Edge-enhancing nodes peer with each other**
- Economic model (flow of money)
 - **Posted prices; brokers coordinating market**
 - **Single point of payment for global service**
- Extensibility yet uniformity of enhancements
 - **Universal execution environment at edge**
 - **Just add standardized service modules**

Who Might Adopt?

- Integrated hypergiants are top of food chain
 - Cloud/app, enhancements, private network
- Others in ecosystem could adopt approach:
 - Enhancements less prone to lock-in
 - Could become broadly supported by host OSes
 - Eventually more featureful than hypergiants
 - Define the enhanced Internet's service model
- Not calling for elimination of private networks, but they might adopt standards

This Is Not A Fairy Tale....

- Everything here is technically possible
 - And entirely backwards-compatible
 - Can start with familiar enhancements
- Incentives line up
 - There are those who have incentive to adopt
 - And this creates incentives for the rest to follow
 - *This is the blueprint for ecosystem change!*
- But research is needed to flesh out vision

Core Research Question

- How to interconnect enhancements
 - Sketched one possibility, with much left undone
 - But there must be far better approaches
- Low probability of adoption, huge impact
 - Worthy of our attention
- Research is about creating possibilities, not ensuring outcomes
 - Without possibilities, change is impossible
 - With possibilities, change is almost impossible

Pre-Meeting Conversation

- Near-consensus: we need level playing field
 - That's the driver of innovation
- Many different opinions on everything else
 - From questioning the severity of the problem to suggesting additional research questions
- But there was interest in this general agenda
 - Go from speculation to demonstrable options
 - Then do the hard work of effecting change....

Additional Research Questions

- Systems/networking research
 - Scalability, resilience, performance, verification, etc.
- Supporting hardware acceleration
 - Design enhancements that can be accelerated
 - Without tying them to specific accelerators
- Network neutrality, economic viability?
 - Theoretical analysis?
- Extent of evolvability?
 - Application-driven networking?

Thank You For Listening!